

Modelling Volatility I: The ARCH Model

Week 2, Day 1 – Advanced Financial Econometrics

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Reference: Tsay, *Analysis of Financial Time Series*, Ch. 3 (Engle, 1982).
All empirical results use the course `data/` folder: S&P 500 daily returns, 2000–2026.

Where we were, and where we go today

- **Weeks so far.** We modelled the *level* of a return – its **conditional mean** $\mathbb{E}[r_t \mid \mathcal{F}_{t-1}]$ – with AR, MA and ARIMA.
- Those models answered: “*given the past, what return do I expect on average tomorrow?*”

Today’s question is different – and, in finance, more important

Not “*which way?*” but “*how much?*” – how large will tomorrow’s move be, in either direction? That size is **volatility**, and it is the raw material of *risk*.

- We will see *why* the linear (mean) models are silent about this,
- build the first model that speaks to it – Engle’s **ARCH** – *step by step*, and
- estimate and forecast it on real S&P 500 data.

Roadmap

Why linear models are not enough

What AR, MA and ARIMA actually model

They all describe the *conditional mean*

$$r_t = \underbrace{\mu_t}_{\text{conditional mean}} + a_t, \quad \mu_t = \mathbb{E}[r_t \mid \mathcal{F}_{t-1}].$$

- **AR**(p): $\mu_t = \phi_0 + \sum_{i=1}^p \phi_i r_{t-i}$ (mean depends on past *returns*).
- **MA**(q): $\mu_t = c + \sum_{j=1}^q \theta_j a_{t-j}$ (mean depends on past *shocks*).
- **ARIMA**(p, d, q): difference d times to reach stationarity, then ARMA the result.

The hidden common assumption

In every one of these models the “error” a_t is treated as **white noise with constant variance**: $\text{Var}(a_t \mid \mathcal{F}_{t-1}) = \sigma^2$ for all t . *The variance is assumed not to move.* That single assumption is what breaks.

Drawback 1 – the mean is almost unpredictable

- For daily S&P 500 returns, the best linear model is a whisper: an AR(1) coefficient of $\hat{\phi}_1 = \text{-0.10}$, explaining almost none of the variation.
- Return *direction* is close to a coin flip – markets are (nearly) efficient in the mean.

So a mean model alone gives you

$$\hat{r}_{t+1} \approx \bar{r} \approx 0, \quad \text{with a fixed } \pm 2\sigma \text{ band around it.}$$

A flat forecast and a flat, unchanging measure of risk. *Useless for risk management.*

- If direction is unpredictable, is there *anything* predictable in returns?
- **Yes – their *size*.** That is where all the structure hides.

Drawback 2 – volatility clusters, but the models assume it can't

Calm huddles with calm; storms huddle with storms (2008, 2020). A constant-variance model draws *one* band across this entire picture – far too wide in quiet years, far too narrow in a crisis. *Real life: turbulence – a wild day makes tomorrow likely wild too.*

The smoking gun: returns are white, but their *size* is not

- **Left:** the ACF of r_t is essentially zero – exactly the “white noise” an ARMA model expects, and the reason the mean is unpredictable.
- **Middle & right:** the ACF of r_t^2 and $|r_t|$ stays large and positive for *dozens* of lags. A big move today predicts a big move tomorrow.

Linear models only look at the left panel. All the predictable structure is in the other two.

Why r_t^2 and $|r_t|$? Measuring *size*, not direction

The trick: throw away the sign, keep the magnitude

Both are **non-negative** and **large exactly when the move is large**, near zero on calm days – they are **volatility proxies**. A +3% day and a -3% day are opposite in *direction* but identical in *size*: $r_t^2 = 9$ and $|r_t| = 3$ for both.

Why r_t^2 is the natural one – it *is* the variance

With (near-)zero mean, $\sigma_t^2 = \text{Var}(r_t \mid \mathcal{F}_{t-1}) = \mathbb{E}[r_t^2 \mid \mathcal{F}_{t-1}]$, so a single r_t^2 is a **one-day estimate of the variance** σ_t^2 (and $|r_t|$ of the volatility σ_t). Hence

autocorrelation in $r_t^2 \equiv$ persistence in the variance $\equiv$ **volatility clustering**,

which is *precisely* what ARCH will model: $\sigma_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2$.

Why show *both*

r_t^2 : direct link to the variance and to ARCH's algebra, but squaring *magnifies outliers* $\Rightarrow$ noisier ACF. $|r_t|$: **more robust**, usually cleaner and stronger persistence (the “Taylor effect”: $|r_t|$ autocorrelates *more* strongly than r_t^2). Agreement across both = the clustering is real, not an artefact of one transform.

Ljung–Box makes it official

Testing “no autocorrelation up to lag 20” on daily S&P 500

Series	$Q(20)$	p -value
Returns r_t	161	$< 10^{-3}$ (weak)
Squared returns r_t^2	8543	≈ 0
Absolute returns $ r_t $	10831	≈ 0

- The dependence in r_t^2 is *two orders of magnitude* stronger than in r_t .
- Uncorrelated returns *do not* mean independent returns: r_t and r_{t-1} are linearly unrelated yet clearly *dependent through their variance*.
- An ARMA model, which sees only linear correlation, is blind to this by construction.

Drawback 3 – fat tails: crashes happen too often for a Normal

- Sample **kurtosis** = 13.7 (Normal = 3); excess = 10.7.
- The peak is too tall, the shoulders too thin, the **tails too fat**.
- The Q–Q points bend away from the line at both ends: extreme days are *far* more frequent than a bell curve allows.

A Gaussian ARMA under-states the chance of a crash – precisely the event a risk manager cares about most.

Even the *best* linear model leaves this untouched

Fit an AR(1) to the returns and inspect its residuals e_t . **Left:** the residual ACF is now (nearly) clean – the mean model did its small job. **Right:** the ACF of the *squared* residuals e_t^2 is as strong as ever. *Ljung-Box*(20) on e_t^2 : $p \approx 0$. Removing mean dependence does nothing to the variance dependence.

Three stylised facts the linear toolkit cannot capture

- ① **No mean predictability:** returns are (near) serially uncorrelated.
- ② **Volatility clustering:** r_t^2 and $|r_t|$ are strongly, persistently autocorrelated.
- ③ **Fat tails:** unconditional returns are far more *leptokurtic* (sharply peaked, with heavy tails) than a Normal.

The gap

AR/MA/ARIMA model the *mean* and assume the variance is a constant σ^2 . But the variance is the part that *moves*, *clusters*, and *drives risk*. We need a model for the **conditional variance** itself.

That model is ARCH. But first: what, precisely, do we mean by “volatility”?

Volatility: the conditional variance

Defining volatility

Volatility = the conditional standard deviation of the return

$$\sigma_t = \sqrt{\text{Var}(r_t \mid \mathcal{F}_{t-1})}, \quad \mathcal{F}_{t-1} = \{r_{t-1}, r_{t-2}, \dots\} \text{ (the past).}$$

- “**Conditional**” is the whole point: given *what we know today*, how uncertain is tomorrow?
- It is **not observed** – we never see σ_t , only the single return r_t it helped generate. We must *model* it.
- It is **time-varying**: today’s σ_t is small in calm markets, large in a crisis.

Two variances – do not confuse them

Unconditional $\text{Var}(r_t)$: one number for the whole sample (a long-run average).

Conditional $\text{Var}(r_t \mid \mathcal{F}_{t-1}) = \sigma_t^2$: a moving quantity, re-evaluated every day. ARMA assumes the second equals a constant. It does not.

The modelling decomposition

Split every return into “where it points” and “how big it is”

$$r_t = \mu_t + a_t, \quad a_t = \sigma_t \varepsilon_t, \quad \varepsilon_t \stackrel{iid}{\sim} (0, 1).$$

- μ_t – the **conditional mean** (last week’s ARMA; for daily returns, ≈ 0).
- $a_t = r_t - \mu_t$ – the **shock** (a.k.a. “mean-corrected return” or innovation).
- σ_t – the **volatility**: the scale that inflates or deflates the shock.
- ε_t – the **standardised innovation**: mean 0, variance 1, i.i.d. – the “pure noise”.

The task of a volatility model

Describe how σ_t^2 evolves given the past. Since $\mathbb{E}[a_t^2 \mid \mathcal{F}_{t-1}] = \sigma_t^2 \mathbb{E}[\varepsilon_t^2] = \sigma_t^2$, the **past squared shocks** a_{t-i}^2 are the natural ingredients. That single idea is ARCH.

The ARCH model, step by step

Engle's idea (1982): let variance feed on past shocks

- **Observation:** a large $|a_{t-1}|$ tends to be followed by a large $|a_t|$ (clustering).
- **Simplest formalisation:** make today's variance an increasing function of *yesterday's squared shock*.

The ARCH(1) model – *AutoRegressive Conditional Heteroskedasticity*

$$\begin{aligned}a_t &= \sigma_t \varepsilon_t, & \varepsilon_t &\stackrel{iid}{\sim} (0, 1), \\ \sigma_t^2 &= \alpha_0 + \alpha_1 a_{t-1}^2.\end{aligned}$$

- “**Heteroskedasticity**” = non-constant variance. “**Conditional**” = given the past. “**Autoregressive**” = the variance regresses on its own recent (squared) history.
- Read it plainly: *a big shock yesterday (a_{t-1}^2 large) $\Rightarrow$ a wide distribution today (σ_t^2 large) $\Rightarrow$ another big shock is likely*. Clustering is built in.

Reading the two equations carefully

$$a_t = \sigma_t \varepsilon_t, \quad \sigma_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2.$$

- σ_t^2 is **known at time $t - 1$** : it is a fixed number once a_{t-1} is observed. Given the past, the *scale* of today's move is *not* random.
- The *direction* is still random and unpredictable: ε_t decides the sign, and $\mathbb{E}[a_t \mid \mathcal{F}_{t-1}] = \sigma_t \mathbb{E}[\varepsilon_t] = 0$.
- So ARCH keeps returns **unpredictable in mean** (fact 1) while making their **size predictable** (fact 2). Exactly the split the data demanded.

Parameter meaning

$\alpha_0 > 0$ – the **baseline** variance (a floor in perfectly calm times).

$\alpha_1 \geq 0$ – the **sensitivity** to shocks: how violently variance reacts to news.

Constraints – what the parameters are *allowed* to be

1. A variance can never be negative

$\sigma_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2 \geq 0$ for *every* possible $a_{t-1} \implies \alpha_0 > 0$ **and** $\alpha_1 \geq 0$.

2. The variance must not explode – (weak) stationarity

For a finite, constant unconditional variance we will need $0 \leq \alpha_1 < 1$ – this is *weak stationarity*: a mean and variance that do not drift with time.

If $\alpha_1 \geq 1$, a shock amplifies itself forever and the process is non-stationary.

3. For finite *fourth* moment (fat-tail formula)

Gaussian ε_t : additionally $3\alpha_1^2 < 1$, i.e. $\alpha_1 < 0.577$.

These are not technicalities – each one is “a variance/kurtosis must exist.”

From ARCH(1) to ARCH(q): a longer memory

- One lag rarely captures the persistence we saw (dozens of significant lags). Let variance react to the *last* q squared shocks.

The ARCH(q) model

$$a_t = \sigma_t \varepsilon_t, \quad \sigma_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2 + \alpha_2 a_{t-2}^2 + \cdots + \alpha_q a_{t-q}^2 = \alpha_0 + \sum_{i=1}^q \alpha_i a_{t-i}^2.$$

- **Constraints:** $\alpha_0 > 0$, $\alpha_i \geq 0$, and $\sum_{i=1}^q \alpha_i < 1$ (stationarity).
- **Unconditional variance:** $\text{Var}(a_t) = \frac{\alpha_0}{1 - \sum_{i=1}^q \alpha_i}$.
- Larger q = longer volatility memory – but the parameter count grows one-for-one with it. For the long memory equities show, that is expensive and fragile; *we return to this cost, and its fix, at the end.*

Properties of the ARCH model

Property 1 – mean zero and no autocorrelation

Work with ARCH(1); ε_t is independent of the past, mean 0, variance 1.

a_t has mean zero

$$\mathbb{E}[a_t] = \mathbb{E}[\sigma_t \varepsilon_t] = \mathbb{E}[\sigma_t] \mathbb{E}[\varepsilon_t] = 0 \quad (\sigma_t \text{ depends only on the past, independent of } \varepsilon_t).$$

a_t is serially *uncorrelated* (white noise)

For $k \geq 1$,

$$\text{Cov}(a_t, a_{t-k}) = \mathbb{E}[a_t a_{t-k}] = \mathbb{E}\left[\underbrace{a_{t-k} \sigma_t}_{\in \mathcal{F}_{t-1}} \varepsilon_t\right] = \mathbb{E}[a_{t-k} \sigma_t] \mathbb{E}[\varepsilon_t] = 0.$$

So ARCH reproduces stylised fact 1: returns look like white noise in the mean.

Property 2 – the unconditional variance

Derivation (law of iterated expectations: $\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid \mathcal{F}_{t-1}]]$)

$$\begin{aligned}\text{Var}(a_t) &= \mathbb{E}[a_t^2] = \mathbb{E}[\mathbb{E}[a_t^2 \mid \mathcal{F}_{t-1}]] = \mathbb{E}[\sigma_t^2] = \mathbb{E}[\alpha_0 + \alpha_1 a_{t-1}^2] \\ &= \alpha_0 + \alpha_1 \mathbb{E}[a_{t-1}^2].\end{aligned}$$

Under stationarity $\mathbb{E}[a_t^2] = \mathbb{E}[a_{t-1}^2] \equiv \sigma^2$, so $\sigma^2 = \alpha_0 + \alpha_1 \sigma^2$, giving

$$\boxed{\text{Var}(a_t) = \sigma^2 = \frac{\alpha_0}{1 - \alpha_1}} \quad (0 \leq \alpha_1 < 1).$$

- This is why we needed $\alpha_1 < 1$: otherwise the denominator is ≤ 0 – no finite variance.
- **Two horizons in one model:** the variance *wobbles* day to day around a fixed long-run level $\alpha_0/(1 - \alpha_1)$. Volatility mean-reverts.

Property 3 – the squares follow an AR(1): this is clustering

A change of variable

Let $\eta_t = a_t^2 - \sigma_t^2$. Then $\mathbb{E}[\eta_t \mid \mathcal{F}_{t-1}] = 0$ (a mean-zero, uncorrelated “surprise”), and substituting $\sigma_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2$:

$$a_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2 + \eta_t.$$

- This is exactly an **AR(1) in a_t^2** with coefficient α_1 .
- Its autocorrelation decays as $\rho_k(a_t^2) = \alpha_1^k$ – **positive and persistent**, matching the middle/right panels of our smoking-gun plot.
- **Punchline:** a_t is white noise, yet a_t^2 is autocorrelated. One model delivers facts 1 and 2 *simultaneously*.

Property 4 – fat tails appear *for free*

Kurtosis of ARCH(1) with Gaussian ε_t (needs $3\alpha_1^2 < 1$)

$$\frac{\mathbb{E}[a_t^4]}{(\mathbb{E}[a_t^2])^2} = 3 \frac{1 - \alpha_1^2}{1 - 3\alpha_1^2} > 3.$$

- Even with *perfectly Gaussian* ε_t , the observed a_t is **leptokurtic**.
- **Intuition:** mixing Normals of different widths (calm days + storm days) piles up mass in the centre *and* in the tails – exactly the fat-tailed shape we measured.
- So ARCH *endogenously* produces stylised fact 3. (*Though, as we will see, for real data it needs help – hence Student-t innovations later.*)

Seeing it: simulate ARCH(1) vs i.i.d. noise

Same standardised noise ε_t drives both panels. **Top** ($\alpha_1 = 0$): plain i.i.d. – shocks evenly sized, no memory.
Bottom ($\alpha_1 = 0.8$): identical randomness, but variance now feeds on past shocks, so quiet stretches and violent bursts emerge on their own. *Clustering is a consequence of the model, not an input.*

Building an ARCH model

The workflow: five honest steps

- ➊ **Model the mean.** Fit an ARMA to r_t (often just a constant), take residuals $a_t = r_t - \hat{\mu}_t$.
- ➋ **Test for ARCH effects.** Are the a_t^2 autocorrelated? (Ljung–Box & Engle’s LM test.)
- ➌ **Choose the order q .** Read the **partial autocorrelation** (PACF) of a_t^2 – it isolates the direct effect of each lag – and confirm with AIC/BIC.
- ➍ **Estimate $\alpha_0, \dots, \alpha_q$ by maximum likelihood.**
- ➎ **Check the fit.** The standardised residuals $\hat{\varepsilon}_t = a_t / \hat{\sigma}_t$ should be i.i.d. with no remaining ARCH.

Same discipline as Box–Jenkins for the mean – now applied to the variance.

Step 2 – testing for ARCH effects: Engle’s LM test

What we are testing

After the mean model, the shocks a_t may be serially *uncorrelated* yet still carry a *moving* variance. The test pits

$$H_0 : \text{Var}(a_t \mid \mathcal{F}_{t-1}) = \sigma^2 \text{ (constant – no ARCH)} \quad \text{vs} \quad H_1 : \text{ARCH}(q).$$

The trick: turn “variance” into a regression on *squared* residuals

Since $\mathbb{E}[a_t^2 \mid \mathcal{F}_{t-1}] = \sigma_t^2$, a single a_t^2 is a **one-day proxy for the variance**. So if the variance feeds on past shocks, then a_t^2 must be *predictable* from its own past. Run the OLS **auxiliary regression**

$$a_t^2 = \beta_0 + \beta_1 a_{t-1}^2 + \cdots + \beta_q a_{t-q}^2 + u_t, \quad H_0 : \beta_1 = \cdots = \beta_q = 0.$$

Under H_0 only the intercept survives – β_0 estimates the constant σ^2 , the lagged squares explain *nothing*, and the fit’s $R^2 \approx 0$. ARCH effects make that R^2 climb.

Engle's LM test – the statistic, and why it works

The statistic: you never have to *fit* ARCH to test for it

Read the R^2 off that single regression and scale it by the sample size:

$$LM = (T - q) R^2 \stackrel{H_0}{\sim} \chi_q^2.$$

- R^2 = the share of a_t^2 's variation explained by its own lags; multiplying by T turns a small-but-nonzero R^2 into a statistic with a known χ^2 law.
- Degrees of freedom = q = the number of restrictions $\beta_1 = \dots = \beta_q = 0$ imposed.
- **“LM” = Lagrange multiplier (score) test:** estimate *only* the cheap null model, then ask whether *adding* the ARCH terms would sharply raise the likelihood. Engle (1982) showed that score collapses exactly to $T \cdot R^2$ – hence the one-regression recipe.

Engle's LM test – the verdict on daily S&P 500

Test on the mean-model shocks a_t (χ_1^2 5% cutoff is only 3.84)

q	$LM = (T-q)R^2$	p -value
1	652	$< 10^{-100}$
5	1666	≈ 0
10	1777	≈ 0

- Even at $q = 1$ the statistic (652) overshoots the 3.84 cutoff by *two orders of magnitude*: the constant-variance null is crushed.
- Adding lags keeps raising LM – the dependence in a_t^2 reaches well beyond one day, an early hint that *one* ARCH lag will not be enough.

Overwhelming ARCH effects – a volatility model is mandatory, not optional.

Step 4 – estimation by maximum likelihood

- OLS cannot fit this: the variance itself is the moving target. We use **MLE**.
- With Gaussian ε_t , the conditional density of a_t given the past is $\mathcal{N}(0, \sigma_t^2)$, so the log-likelihood is

$$\ell(\alpha) = -\frac{1}{2} \sum_t \left[\log 2\pi + \log \sigma_t^2(\alpha) + \frac{a_t^2}{\sigma_t^2(\alpha)} \right], \quad \sigma_t^2(\alpha) = \alpha_0 + \alpha_1 a_{t-1}^2.$$

- Read the penalty: a day with a big a_t^2 but a small predicted σ_t^2 is punished hard – MLE inflates σ_t^2 exactly where big shocks occur.
- Maximise numerically (respecting $\alpha_0 > 0$, $0 \leq \alpha_1 < 1$).

ARCH(1) fitted to daily S&P 500 (MLE)

$$\hat{\sigma}_t^2 = \underset{(\hat{\alpha}_0)}{0.94} + \underset{(\hat{\alpha}_1)}{0.39} a_{t-1}^2, \quad \frac{\hat{\alpha}_0}{1 - \hat{\alpha}_1} = 1.54 \Rightarrow \hat{\sigma} = 1.24\% \text{ per day.}$$

Implied long-run volatility 1.24% matches the sample s.d. 1.22% – a good sign.

The fitted volatility, laid over the returns

The red $\pm 2\hat{\sigma}_t$ band **breathes** with the market: it widens through 2008 and 2020 and contracts in calm years – the single feature a constant-variance ARMA can never reproduce. This band is, in effect, a daily 95% *Value-at-Risk* (VaR) envelope – the move it would take a 1-in-20 day to exceed.

Step 5 – standardised residuals: the idea and the arithmetic

Definition: divide each shock by the volatility the model predicted for that day

$$\hat{\varepsilon}_t = \frac{a_t}{\hat{\sigma}_t}, \quad \hat{\sigma}_t = \sqrt{\hat{\alpha}_0 + \hat{\alpha}_1 a_{t-1}^2} \quad (\text{fixed once } a_{t-1} \text{ is seen}).$$

If the model caught the moving scale, what is left is **scale-free**: it should be i.i.d. with mean 0 and variance 1 – the “pure noise” ε_t we posited in $a_t = \sigma_t \varepsilon_t$.

The *same* -3% day, judged against its own $\hat{\sigma}_t$ (fit $\hat{\sigma}_t^2 = 0.94 + 0.39 a_{t-1}^2$)

Yesterday's shock a_{t-1}	today's $\hat{\sigma}_t$	today's $a_t = -3\%$ standardises to
-4% (a storm)	$\sqrt{0.94 + 0.39(16)} = 2.68\%$	$\hat{\varepsilon}_t = -3/2.68 = -\mathbf{1.1}$ (ordinary)
$+0.3\%$ (calm)	$\sqrt{0.94 + 0.39(0.09)} = 0.99\%$	$\hat{\varepsilon}_t = -3/0.99 = -\mathbf{3.0}$ (extreme)

Three adequacy checks on $\hat{\varepsilon}_t$: (1) mean ≈ 0 , sd ≈ 1 ; (2) Ljung–Box on $\hat{\varepsilon}_t$ – mean structure gone? (3) Ljung–Box on $\hat{\varepsilon}_t^2$ and ARCH-LM – volatility structure gone?

Step 5 – the verdict on our ARCH(1)

Diagnostics on the standardised residuals $\hat{\varepsilon}_t = a_t/\hat{\sigma}_t$

Check	Value	Reading
mean, sd	0.00, 1.00	centred and unit-scaled ✓
Ljung–Box $\hat{\varepsilon}_t$	$p = 5 \times 10^{-4}$	faint leftover mean dependence
Ljung–Box $\hat{\varepsilon}_t^2$	$p \approx 0$	ARCH still remains
kurtosis $\hat{\varepsilon}_t$	10.3 (Normal = 3)	tails still fat

Two honest lessons – and two fixes

- (a) One lag is too short; daily equity volatility has long memory. A high-order ARCH(q) or – far more parsimoniously – **GARCH** (next lecture) is needed.
- (b) Gaussian innovations under-state the tails even after ARCH. The fix is a heavier-tailed ε_t : the **Student- t** .

Does a longer memory help? ARCH(1) vs (2) vs (5), S&P 500

Refit at higher orders; re-run the Step-5 diagnostics on $\hat{\varepsilon}_t$

Order	Persistence $\sum_i \hat{\alpha}_i$	Leftover kurtosis	Ljung–Box $\hat{\varepsilon}_t^2$	ARCH-LM ₁₀ ($\hat{\varepsilon}_t$)
ARCH(1)	0.39	10.3	≈ 0	$\sim 10^{-217}$
ARCH(2)	0.61	6.5	≈ 0	$\sim 10^{-70}$
ARCH(5)	0.82	4.7	9×10^{-10}	4×10^{-9}

- **Persistence climbs toward 1** (0.39 $\rightarrow$ 0.61 $\rightarrow$ 0.82): the true volatility memory is *long* – one lag saw only a slice of it.
- **Leftover ARCH collapses** (LM exponent $-217 \rightarrow -9$) and kurtosis falls $10.3 \rightarrow 4.7$: each added lag mops up more clustering and more of the tail.
- **But it never clears** – even five lags leave significant ARCH. Matching this memory with pure ARCH would need *many* α 's: **exactly the inefficiency GARCH removes** with a single β .

Student- t innovations

Why Gaussian innovations are not enough

- ARCH manufactures *some* excess kurtosis by mixing variances – but for daily equities, *not enough*.
- Our standardised residuals $\hat{\varepsilon}_t$ still have kurtosis $\approx 10 \gg 3$: the model explains the *clustering* of extremes but not their *frequency*.

The remedy: give the noise fatter tails to begin with

Replace $\varepsilon_t \sim \mathcal{N}(0, 1)$ with a **standardised Student- t** :

$$\varepsilon_t = \frac{t_\nu}{\sqrt{\nu/(\nu - 2)}}, \quad \nu > 2, \quad \mathbb{E}[\varepsilon_t] = 0, \quad \text{Var}(\varepsilon_t) = 1.$$

(We rescale so the variance is still 1 – all the “size” stays in σ_t .)

The degrees of freedom ν is the single tail dial – next we unpack exactly what it is.

What the degrees-of-freedom parameter ν actually is

ν is the Student- t 's *shape* parameter – the one knob for tail heaviness

It is not a mean or a variance: those are already handled by μ_t and σ_t . “Degrees of freedom” historically counts independent pieces of information; here it survives as a pure **shape** number – the *smaller* ν , the *heavier* the tails (more frequent extremes).

Why it is relevant: ν decides *which moments even exist*

A Student- t_ν has a finite moment of order k only when $k < \nu$:

mean exists $\Leftrightarrow \nu > 1$, variance exists $\Leftrightarrow \nu > 2$, kurtosis exists $\Leftrightarrow \nu > 4$.

So we need $\nu > 2$ merely to rescale to variance 1; and the excess-kurtosis formula $\frac{6}{\nu - 4}$ holds *only* for $\nu > 4$ – below that the fourth moment is infinite.

- **Our $\hat{\nu} \approx 3.2$:** the variance exists (> 2) but the *fourth moment does not* (< 4). That is why the sample kurtosis (13.7) is so large and unstable – it estimates something the population does not possess.
- $\nu \rightarrow \infty$: every moment finite, tails vanish, Student- $t \rightarrow$ Normal. Small ν : violent extremes.

Estimating ν , and checking the tails

- ν is estimated *jointly* with α_0, α_1 by MLE – just swap the Student- t density for the Normal in the likelihood; the ARCH recursion for σ_t^2 is unchanged.
- **On our S&P 500 data:** $\hat{\nu} \approx 3.2$, matching the heavy sample kurtosis of 13.7.

Left: standardised residuals bend off the Normal line at both tails (too many extremes). **Right:** against a Student- $t(\hat{\nu})$ they fall on the line – the tails now fit.

Does it travel? The same three facts on Apple and Microsoft

ARCH(1) and ARCH(5) fitted to daily AAPL and MSFT log-returns

Asset	Order	Persistence $\sum \hat{\alpha}_i$	Long-run $\hat{\sigma}$ (%/day)	Leftover kurt.	LB $\hat{\varepsilon}_t^2$
AAPL	ARCH(1)	0.48	2.60	10.1	≈ 0
AAPL	ARCH(5)	0.81	2.94	9.3	1×10^{-6}
MSFT	ARCH(1)	0.39	1.98	10.9	≈ 0
MSFT	ARCH(5)	0.75	2.12	9.4	0.23

- **Single stocks out-move the index:** long-run $\hat{\sigma} \approx 2\text{--}3\%$ /day vs 1.24% for the S&P 500 – diversification calms the index, not its members.
- **Same signature everywhere:** persistence again climbs with q ; clustering is no index artefact.
- **One finally clears:** MSFT's ARCH(5) passes the $\hat{\varepsilon}_t^2$ test ($p = 0.23$) – five lags suffice for it, while AAPL and the S&P still show leftover ARCH.
- **Tails are universally heavy:** Student- t $\hat{\nu} \approx 3.2$ (S&P), 3.4 (AAPL), 3.3 (MSFT).

Forecasting volatility

Forecasting variance – the payoff of the whole exercise

Stand at time t (origin), forecast the variance h steps ahead.

One step ahead – and it is *known*

$$\sigma_t^2(1) = \mathbb{E}[a_{t+1}^2 \mid \mathcal{F}_t] = \alpha_0 + \alpha_1 a_t^2.$$

Because a_t is already observed, tomorrow's variance carries *no* forecast error.

h steps ahead – recurse, using $\mathbb{E}[a_{t+h-1}^2 \mid \mathcal{F}_t] = \sigma_t^2(h-1)$

$$\sigma_t^2(h) = \alpha_0 + \alpha_1 \sigma_t^2(h-1), \quad h = 2, 3, \dots$$

- Solving the recursion: $\sigma_t^2(h) - \sigma^2 = \alpha_1^{h-1}(\sigma_t^2(1) - \sigma^2)$, where $\sigma^2 = \frac{\alpha_0}{1 - \alpha_1}$.
- So the forecast **mean-reverts geometrically** to the long-run variance σ^2 at rate α_1 .

The forecast in one picture

From a quiet origin, the 1-day forecast is below the long-run level (0.98%); each further step is pulled toward $\sigma = 1.24\%$ (dashed), reaching it within $\sim$ two weeks.

- From a *turbulent* origin the curve would start *above* the line and decay down to it.
- **Economic reading:** volatility shocks are temporary – markets calm down, but not instantly.

From variance forecast to a return interval

- A variance forecast becomes a **risk statement** through the innovation distribution:

$$r_{t+1} \in \hat{\mu}_{t+1} \pm q_{1-\alpha/2} \hat{\sigma}_t(1),$$

with q the quantile of ε_t – 1.96 if Gaussian, but **larger under Student- t** .

- **Why the choice of innovation matters:** with $\hat{\nu} \approx 3.2$ the 99% quantile is well beyond 2.33, so the t -based interval is wider – a more honest tail, and a higher Value-at-Risk.

What ARCH buys the risk manager

Earlier the *fitted* band moved with the market; now the *forecast* band does too – its width set by $\hat{\sigma}_t(h)$, its reach by the innovation quantile q . Gaussian $q = 1.96$ is optimistic in the tail; the wider Student- t quantile is what makes the Value-at-Risk honest.

Bringing it together: a worked exercise

Exercise – an ARCH(1) desk model

Setup

A trading desk models daily returns (in %) as a zero-mean ARCH(1):

$$r_t = a_t, \quad a_t = \sigma_t \varepsilon_t, \quad \varepsilon_t \stackrel{iid}{\sim} (0, 1), \quad \sigma_t^2 = 0.6 + 0.4 a_{t-1}^2 \quad (\text{units: } \%^2).$$

Yesterday was a turbulent session: the shock was $a_t = +2\%$.

Your tasks

- (a) Is the model **stationary**? State the condition and check it.
- (b) Find the **long-run** (unconditional) variance and daily volatility.
- (c) Forecast **tomorrow's** volatility σ_{t+1} . Calmer or stormier than average – and why?
- (d) Forecast **two days** ahead, $\sigma_t^2(2)$.
- (e) To **what level**, and **how fast**, do the h -day forecasts converge?
- (f) Give an approximate **95% range** for tomorrow's return (Gaussian ε).
- (g) *Stretch*: with Gaussian ε , what is the **kurtosis** of a_t ?

Try (a)–(c) now; then we build (d)–(g) together.

Solution – parts (a)–(c)

(a) Stationarity. Need $\alpha_0 > 0$, $\alpha_1 \geq 0$, $\alpha_1 < 1$. Here $\alpha_1 = 0.4 < 1$ ✓ – a finite unconditional variance exists.

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(b) Long-run level.

$$\sigma^2 = \frac{\alpha_0}{1 - \alpha_1} = \frac{0.6}{1 - 0.4} = \frac{0.6}{0.6} = 1.0 \%^2 \implies \sigma = 1.0\% \text{ per day.}$$

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(c) One day ahead. σ_{t+1}^2 is *known* once a_t is observed:

$$\sigma_{t+1}^2 = \alpha_0 + \alpha_1 a_t^2 = 0.6 + 0.4 (2)^2 = 0.6 + 1.6 = 2.2 \%^2 \implies \sigma_{t+1} = \sqrt{2.2} \approx \mathbf{1.48\%}.$$

1.48% > 1.0%: **stormier than average**. Yesterday's big shock feeds forward – *volatility clustering*.

Solution – parts (d)–(g)

(d) Two days ahead, using $\mathbb{E}[a_{t+1}^2 \mid \mathcal{F}_t] = \sigma_{t+1}^2$:

$$\sigma_t^2(2) = \alpha_0 + \alpha_1 \sigma_{t+1}^2 = 0.6 + 0.4(2.2) = 1.48 \text{ \%}^2 \Rightarrow \sigma \approx 1.22\% \quad (\text{already falling toward } 1.0).$$

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(e) Convergence. To the unconditional $\sigma^2 = 1.0 \%^2$, **geometrically at rate $\alpha_1 = 0.4$** :

$$\sigma_t^2(h) - 1 = 0.4^{h-1}(\sigma_{t+1}^2 - 1) = 0.4^{h-1}(1.2) \quad (\text{gap shrinks } \times 0.4 \text{ per day} \Rightarrow \text{calm within a week}).$$

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(f) 95% range for tomorrow (Gaussian, mean 0):

$$0 \pm 1.96 \sigma_{t+1} = \pm 1.96(1.48) \approx \pm \mathbf{2.9\%} \quad (\text{vs } \pm 1.96\% \text{ in normal times – the band widened}).$$

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(g) Kurtosis (needs $3\alpha_1^2 = 0.48 < 1$ ✓):

$$\frac{\mathbb{E}[a_t^4]}{(\mathbb{E}[a_t^2])^2} = 3 \frac{1 - \alpha_1^2}{1 - 3\alpha_1^2} = 3 \frac{1 - 0.16}{1 - 0.48} = \frac{2.52}{0.52} \approx \mathbf{4.85} > 3 \quad (\text{fat tails, even Gaussian } \varepsilon).$$

What the exercise showed

One model, five payoffs – all from two equations and one shock

- **Stationarity** ($\alpha_1 < 1$) guarantees a finite long-run variance $\alpha_0/(1 - \alpha_1)$.
- **Tomorrow's risk is known today**: $\sigma_{t+1}^2 = \alpha_0 + \alpha_1 a_t^2$ – a big shock $\Rightarrow$ a wide band.
- **Forecasts mean-revert** to the long-run level, geometrically at rate α_1 .
- The **95% band breathes**: $\pm 2.9\%$ after a storm vs $\pm 1.96\%$ in calm times.
- **Fat tails** emerge unbidden: here the kurtosis works out to $4.85 > 3$, with Gaussian ε .

Every number came straight from $\sigma_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2$ and yesterday's a_t .

Today in one slide

The problem

- AR/MA/ARIMA model the *mean*; they assume constant variance.
- Data say otherwise: returns are white, but r_t^2 clusters and tails are fat.

The model: ARCH(q)

- $a_t = \sigma_t \varepsilon_t$, $\sigma_t^2 = \alpha_0 + \sum_{i=1}^q \alpha_i a_{t-i}^2$.
- Constraints: $\alpha_0 > 0$, $\alpha_i \geq 0$, $\sum \alpha_i < 1$.

Properties

- Mean 0, *no* autocorrelation in a_t ; AR structure in a_t^2 .
- Unconditional var = $\alpha_0/(1 - \alpha_1)$; endogenous fat tails.

Practice

- Test: Ljung–Box on a_t^2 , Engle LM. Fit by MLE.
- Student- t innovations for the remaining tails.
- Variance forecast mean-reverts to $\alpha_0/(1 - \alpha_1)$.

The empirical thread

Daily S&P 500: kurtosis 13.7, ARCH-LM $p \approx 0$, $\hat{\alpha}_1 \approx 0.39$, $\hat{\nu} \approx 3.2$. ARCH(1) captures the *mechanism* but needs many lags – **next: GARCH**.

Check your understanding

- ① Returns are serially uncorrelated but their squares are strongly autocorrelated. Which stylised fact is this, and which model class does it motivate?
- ② In $\sigma_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2$, why must $\alpha_1 < 1$? What breaks if $\alpha_1 = 1.2$?
- ③ ARCH keeps $\mathbb{E}[a_t | \mathcal{F}_{t-1}] = 0$ yet makes $\text{Var}(a_t | \mathcal{F}_{t-1})$ time-varying. In words, what is predictable and what is not?
- ④ You fit an ARCH(1); the squared standardised residuals are *still* autocorrelated. What does this tell you, and what would you try next?
- ⑤ Why prefer Student-*t* over Gaussian innovations, even though ARCH already generates excess kurtosis?

(Answers: volatility clustering $\rightarrow$ ARCH/GARCH; stationarity/finite variance, it explodes; the size of the move is predictable, the direction is not; one lag is too short $\rightarrow$ higher order or GARCH; ARCH's kurtosis is too small for real tails.)

Next time

From ARCH to GARCH

ARCH(q) needs many lags to match the long memory in volatility – costly and fragile. **GARCH** lets σ_t^2 depend on its *own* past as well as past shocks:

$$\sigma_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2 + \beta_1 \sigma_{t-1}^2,$$

capturing the same persistence with a handful of parameters. *That is Day 2.*

Questions?

Appendix – reproduce everything in R

All figures and numbers come from `Day1_ARCH_code.R` on the local `data/` folder.

```
r <- 100*diff(log(idx$SP500))    # S&P 500 daily log returns (%)
a <- r - mean(r)                 # mean-adjusted shocks

# ARCH effects: Ljung-Box on squares + Engle's LM test
Box.test(a^2, lag = 20, type = "Ljung-Box")
# LM: regress a_t^2 on q lags -> (T-q)*R^2 ~ chi^2(q)

# Fit ARCH(1), Gaussian and Student-t, by MLE (fGarch)
library(fGarch)
garchFit(~garch(1,0), data=a, cond.dist="norm", include.mean=FALSE)
garchFit(~garch(1,0), data=a, cond.dist="std", include.mean=FALSE)

# Forecast:  $\sigma^2(h) = a_0 + a_1 \sigma^2(h-1) \rightarrow a_0/(1-a_1)$ 
```

Fitted (daily S&P 500): $\hat{\alpha}_0 = 0.94$, $\hat{\alpha}_1 = 0.39$, $\hat{\nu} \approx 3.2$; long-run $\hat{\sigma} = 1.24\%$.